

M2.11 Team Deliverable 1: Waterfall Plan

Smart Factory Team

Team Members: Hoang Nguyen, William Miller, Chase Post, Shiloh Parker, Thierry Tran

Instructions:

To receive credit for your progress, please attach the following documentation.

1. Products Requirement Document PDF.
2. A link to your team's GitHub repository.
3. Please invite your instructors(s) to your repository to make sure they have access.
My github id is davidalanreid.
4. A PDF or Word copy of your Scrum documents:
 - a. Scrum minutes—This is your meeting documentation.
 - b. Product backlog—For Module 2, this should include your waterfall plan and the overall concept of the idea at the top.

Github Repository: <https://github.com/SmartFactory-Team>

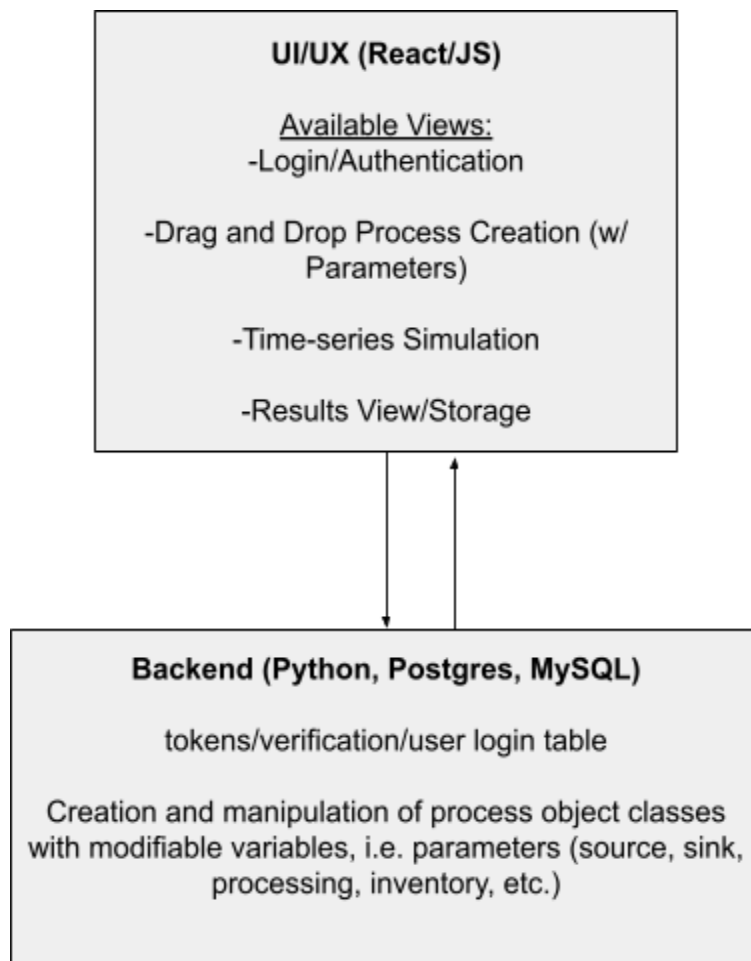
Product Vision:

We envision a cloud-based smart factory application that allows users to model, simulate, and optimize factory operations. For our MVP, we will focus on modeling an aluminum coil plant. We prioritize robust security features, production parameter optimization, and simulation/reporting functionality. We plan to build the UI/UX with React/JS and the backend with Python and Postgres/MySQL databases.

Competitive Advantage / Success Criteria:

The envisioned product competes with AnyLogic, Flexsim, and other desktop applications designed to model and simulate real-world operational systems. The envisioned product benefits from a modernized UI/UX, a cloud native, scalable architecture (SaaS), and more effective actionable reporting for executive stakeholders. For the purposes of this class, our goal is to deliver an MVP that demonstrates the idea with an aluminum coil plant example.

Architecture Diagram:



Draft of Requirements / Product Backlog:

As a production analyst, I'd like the ability to model and optimize an aluminum production process, so I can run a line that maximizes profit.

As a production analyst, I'd like the ability to run simulations and see reports to understand the dynamics of our aluminum production process.

As an IT engineer, I'd like the ability to easily view, modify, and maintain the code for this application, so I can keep my end-users satisfied.

As a security analyst, I'd like the application to be compliant with at least 1 cybersecurity framework, so I can mitigate cyber risk.

As a production controller, I'd like the ability to connect the cloud-based production model to real-time, real-world data feeds, so I can have a digital twin that supports my decision-making.

Waterfall Plan:

With the above product backlog in mind, we define high-level phases below. Agile execution (via sprints) will be used with continuous integration and delivery efforts in each sprint.

Requirements Ideation and Refinement (2/2 - 2/18 OR 2/2 - 2/25, as assigned)

- Articulate epics to form product backlog
- Rank epics by priority/dependencies, breaking the first 1-2 epics into user stories.

Development Sprints (2/19 - 4/9 OR 2/26 - 4/16, as assigned)

- Iterative development in 3 sprints, each 2 weeks long.
 - Sprints last 14 days, starting every other Thursday
 - Every other Thursday's standup initiates a sprint
- Sprints will align with the course schedule and begin either February 19 or February 26, as assigned.
- Conduct a comprehensive code review at least once every 4 weeks
 - 3/12 and 4/16 tentative dates.
- Test and review the iterative product with the customer regularly, ideally at the end of each sprint.

Final Integration and Delivery (4/10 - 4/23 OR 4/17 - 4/30)

- Final User Acceptance Testing to validate quality and confirm delivery.

Final Presentation (4/23 or 4/30, as assigned)

Scrum Reports:

Meeting time: February 12th, 2026

Present for the Meeting (All Team Members)

- Hoang Nguyen
- William Miller
- Chase Post
- Shiloh Parker
- Thierry Tran

Completed Actions

- Finished the Waterfall Plan Team Deliverable.
- Decided next meeting date.

Decisions / Assigned Tasks

- We decided as a group that we prefer being in Section 1 (2/19 start date)
- Individually brainstorm survey questions and align over the weekend.
- Finish and populate needfinding report/survey by Thursday 2/19, when sprint 1 begins.

Summary / TO-DO

- Create a series of questions to gather information for a Needfinding Report (preferably 6-7 questions)

Meeting time: February 5th, 2026

Present for the Meeting (All Team Members)

- Hoang Nguyen
- William Miller
- Chase Post
- Shiloh Parker
- Thierry Tran

Completed Actions

- Created a repository for the project.
- Created a Discord server for communicating among members.
- Decided next meeting date.

Decisions / Assigned Tasks

- We discussed a plan to collaborate on forming a high-level statement of our project objective/vision. **Done**
- We are considering drawing an architecture diagram or similar to visualize the product design at a high-level. **Done**
- We will then write user stories to define the core product functionality (our product backlog). **Done**
- We will build a waterfall schedule to show how we might approach the project with the waterfall methodology. **Done**
- We will specify that we plan to instead take an agile approach with weekly scrum meetings (rotating scrum master) to review progress, address blockers, and prepare for sprint reviews with the customer. **Done**

Summary / TO-DO

- We agreed to iterate independently on the requirements, diagram, waterfall plan on February 6th before aligning and refining our collective document on February 7th. **Done**

Aluminum Coil Plant - Process Overview:

First, the process starts with recycled cans and pure aluminum blocks and melts them down into ingots. These are usually done in batches of 4-9 but it tries to use as much recycled material as possible to save on cost. The pure aluminum is then used to counteract any issues in the material like foreign elements and such to course-correct the alloy back to the desired chemistry.

Pure Aluminum blocks



Recycled Can



Unscalped Aluminum Ingot



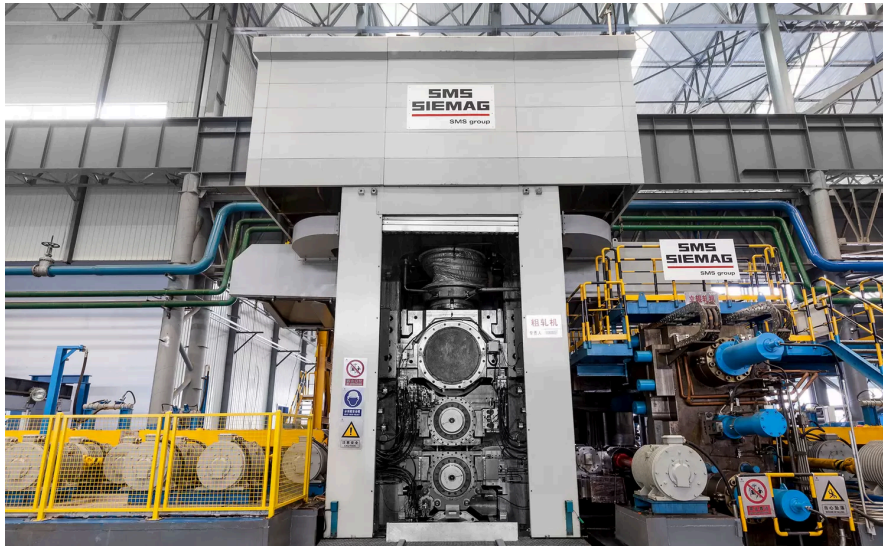
After the ingot has been cast, it is sent to a scalper for cleaning. The scalper's job is to cut the minimum amount of material off of the top and bottom of an ingot to create a smooth surface to run the ingot through the mill.

Scalped Ingot (notice the smooth clean top)



An aluminum mill contains what are called stands and a stand is a set of rolls that are working the aluminum down to a certain thickness. These can have 2,4, or 6 rolls per stand and mills can have 1 to 8 stands usually. If they have more than 2 rolls, these are divided into two categories, work rolls and backup rolls. The work rolls are smaller and are the ones that are actually touching the metal, the backup rolls are much larger and are there to generate the main force behind the milling process. The way they work is the gap between the rolls gets smaller as the metal passes through which decreases the thickness but increases the length of the metal until it is thin enough to turn into a coil.

Example of a 1 stand mill (This one has 4 rolls total)



The aluminum ingot after milling, rolled into a coil.

